

The reactivity of three oxides mixed in H_2O , HCl , and NaOH is shown in the table. All experiments were performed at $25\text{ }^\circ\text{C}$.

Oxide	Mixed with $\text{H}_2\text{O}(\text{l})$	Mixed with $\text{HCl}(\text{aq})$	Mixed with $\text{NaOH}(\text{aq})$
$\text{SO}_2(\text{g})$	aqueous solution, $\text{pH} < 7$	no reaction	salt forms
$\text{MgO}(\text{s})$	aqueous solution, $\text{pH} > 7$	salt forms	no reaction
$\text{N}_2\text{O}(\text{g})$	no reaction, $\text{pH} = 7$	no reaction	no reaction

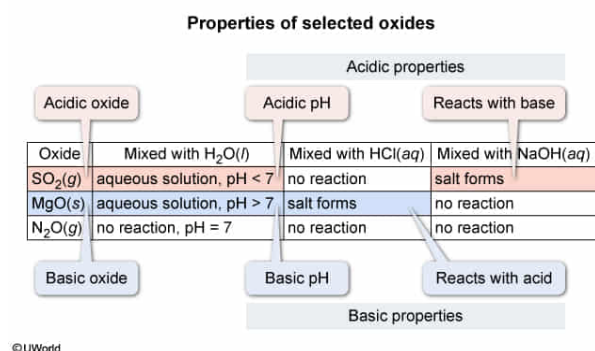
Based on the data, which statement best describes how SO_2 compares to MgO ?

- ☐ A. Both SO_2 and MgO are acidic oxides.
- ☐ B. Both SO_2 and MgO are basic oxides.
- ☐ C. SO_2 is a basic oxide but MgO is an acidic oxide. [8%]
- ✓ ☒ D. SO_2 is an acidic oxide but MgO is a basic oxide. [90%]

Correct

90% answered correctly

Explanation:



Oxides are binary compounds consisting of oxygen bound with only one other element. They can be classified as acidic, basic, or neutral, depending on their reactions with water, acids, and bases. Acidic oxides react with water to form acidic compounds that produce a solution $\text{pH} < 7$. Acidic oxides also react with bases to form salts. In contrast, basic oxides react with water to form basic compounds that yield a solution $\text{pH} > 7$, and basic oxides form salts by reacting with acids. Neutral oxides are unreactive toward water, acids, and bases.

Evaluation of the given data is required to form a conclusion about the compounds. The table shows that SO_2 forms an acidic aqueous solution ($\text{pH} < 7$) and that it reacts with NaOH (a base) to form a salt (similar to an acid-base neutralization). As such, SO_2 exhibits **acidic** properties. Conversely, MgO forms a basic aqueous solution ($\text{pH} > 7$) and reacts with HCl (an acid) to form a salt, so MgO exhibits **basic** properties.

Therefore, SO_2 is an acidic oxide, whereas MgO is a basic oxide.

(Choices A and B) SO_2 and MgO form aqueous solutions with pH values that are on opposite sides of the pH scale relative to pH 7 (a neutral solution). SO_2 and MgO also exhibit the opposite reactivity toward HCl and NaOH . SO_2 and MgO cannot both be the same type of oxide.

(Choice C) SO_2 exhibits *acidic* properties (aqueous solution $\text{pH} < 7$ and reacts with NaOH), whereas MgO exhibits *basic* properties (aqueous solution $\text{pH} > 7$ and reacts with HCl).

Educational objective:

Compounds that react with bases and produce solutions with $\text{pH} < 7$ can be classified as acidic, whereas compounds that react with acids and form solutions with $\text{pH} > 7$ can be classified as basic.

Time spent:0 sec

Subject:

General Chemistry

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